Consolidation and transition

Lecture and worked examples

Maths Explorer

Review of calculation, functions and geometry

1. Reduce $168/252$ to lowest terms; calculate $2^{-3} + 1/4$.
2. Factorise $x^2 - 6x + 9$, then solve $(x-3)^2 = 4$.
3. Solve $\dfrac{x-2}{x+1} \geq 0$.
4. For $f(x) = 4 - (x-1)^2$ on $[-2, 3]$, give the maximum and the abscissae where $f(x) = 0$.
5. With $A(0,1)$, $B(4,3)$, $C(6,4)$, prove collinearity and give an equation of the line.

Worked answer: Review of calculation, functions and geometry

1. $168/252 = 2/3$, and $2^{-3} + 1/4 = 1/8 + 2/8 = 3/8$. 2. $(x-3)^2$; $x - 3 = \pm 2$, so $S = \{1, 5\}$. 3. Forbidden -1, zero 2; signs $+, -, +$, so $S = (-\infty, -1) \cup [2, +\infty)$. 4. The square is positive or zero, so $f(x) \leq 4$, with equality at $x = 1$. The zeros are $x = -1, 3$, both in the domain. 5. $\overrightarrow{AB} = (4, 2)$, $\overrightarrow{AC} = (6, 3) = \frac{3}{2}\overrightarrow{AB}$. An equation is $x - 2y + 2 = 0$.

Review of data, probability and modelling

1. Calculate the mean and median of $2; 4; 4; 6; 9$. Determine Q_1, Q_3.
2. A price drops by 15%, then rises by 20%. Give the overall rate and the rate that then restores the initial price.
3. In a fictional population of 100 people, 40 play sport (S), 30 come by bike (V), 18 do both. Construct the two-way table and calculate $P_S(V)$, $P_V(S)$ for an equally likely draw.

Review of data, probability and modelling (2)

4. For a rectangle of perimeter 24 m, express the area as a function of one dimension x and prove its maximum.

Worked answer: Review of data, probability and modelling

1. Mean $25/5 = 5$, median 4. Quartile ranks 2 and 4, so $Q_1 = 4$, $Q_3 = 6$.

2. Overall multiplier $0.85 \times 1.20 = 1.02$: an increase of 2%. Reverse rate $1/1.02 - 1 = -1/51 \approx -1.96\%$.

3. Table: $S \cap V = 18$, $S \cap \overline{V} = 22$, $\overline{S} \cap V = 12$, $\overline{S} \cap \overline{V} = 48$. $P_S(V) = 18/40 = 45\%$, $P_V(S) = 18/30 = 60\%$.

4. $A(x) = x(12 - x) = 36 - (x - 6)^2$ on $(0, 12)$. The maximum is 36 m^2, attained for the square of side 6 m.

Choosing a strategy and preparing what comes next

For a given function f, explain the following program. What does it return with $f(x) = x^2$?

Worked answer: Choosing a strategy and preparing what comes next

It returns a list of abscissa-image pairs for the integers from -3 to 3 inclusive: $(-3, 9), (-2, 4), (-1, 1), (0, 0), (1, 1), (2, 4), (3, 9)$. The bound 4 of `range` is excluded. The table does not describe the images of all real numbers between -3 and 3.

Python

```
def tableau(f):
    valeurs = []
    for x in range(-3, 4):
        valeurs.append((x, f(x)))
    return valeurs
```

Choosing a strategy and preparing what comes next

Search: try things out, organise leads. **Model**: choose variables, domain and assumptions. **Represent**: move from text to table, figure or graph. **Reason**: justify and distinguish conjecture from proof. **Calculate**: choose a form and check the operations. **Communicate**: explain clearly with suitable notation and units.

Choosing a strategy and preparing what comes next

Choose a successful piece of work and a corrected one. For each, identify the skills used, explain one method and state a personal goal for the following year. Then redo an exercise matching this goal without looking at the correction.

Worked answer: Choosing a strategy and preparing what comes next

There is no single numerical answer. A satisfactory piece of work precisely identifies the problem, the chosen method, a justification and a check. The goal must be observable, for example: “state the forbidden values before solving a quotient equation”, rather than “get better at calculation”.